

Suite Hard · Simulation Module 14

22 questions · 35 minutes · official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

Read the prompt carefully. Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 6

Advanced Math: 5

PSDA: 5

Geometry: 6

Do not open the next module until this one is timed and scored.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

The table below shows the distribution of US states according to whether they have a state-level sales tax and a state-level income tax.

2013 State-Level Taxes

	State sales tax	No state sales tax
State income tax	39	4
No state income tax	6	1

To the nearest tenth of a percent, what percent of states with a state-level sales tax do not have a state-level income tax?

Answer

- A. 6.0%
- B. 12.0%
- C. 13.3%
- D. 14.0%

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question
One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

To determine the mean number of children per household in a community, Tabitha surveyed 20 families at a playground. For the 20 families surveyed, the mean number of children per household was 2.4. Which of the following statements must be true?

Answer

- A. The mean number of children per household in the community is 2.4.
- B. A determination about the mean number of children per household in the community should not be made because the sample size is too small.
- C. The sampling method is flawed and may produce a biased estimate of the mean number of children per household in the community.
- D. The sampling method is not flawed and is likely to produce an unbiased estimate of the mean number of children per household in the community.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

Answer

- A. $(x + a)^2$
- B. $(x - a)^2$
- C. $(x + a)(x - a)$
- D. $x^2 - ax - a^2$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

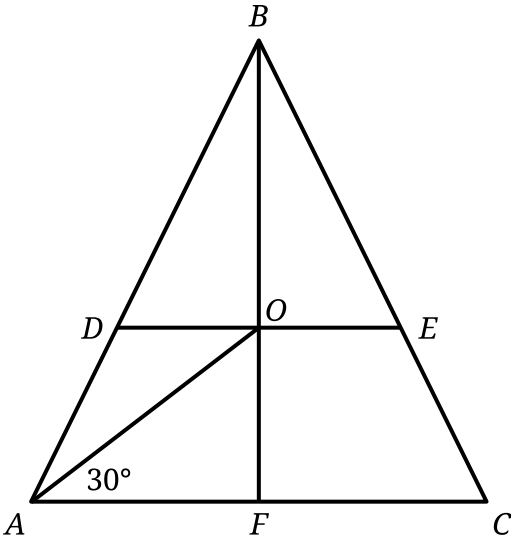
The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by **2.10** kelvins, by how much did the temperature increase, in degrees Fahrenheit?

Answer

- A. **3.78**
- B. **35.78**
- C. **487.89**
- D. **519.89**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

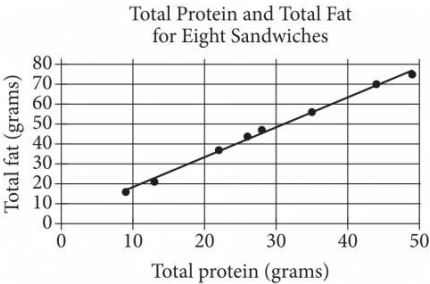


Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Hard

Question



The scatterplot above shows the numbers of grams of both total protein and total fat for eight sandwiches on a restaurant menu. The line of best fit for the data is also shown. According to the line of best fit, which of the following is closest to the predicted increase in total fat, in grams, for every increase of 1 gram in total protein?

Answer

- A. 2.5
- B. 2.0
- C. 1.5
- D. 1.0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

- I. $g(x) = 18(1.16)(1.4)^{x+2}$
- II. $h(x) = 18(1.4)^{x+4}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A manufacturing plant makes **10**-inch, **9**-inch, and **7**-inch frying pans. During a certain day, the number of **10**-inch frying pans that the manufacturing plant makes is **4** times the number ***n*** of **9**-inch frying pans it makes, and the number of **7**-inch frying pans it makes is **10**. During this day, the manufacturing plant makes **100** frying pans total. Which equation represents this situation?

Answer

- A. $10(4n) + 9n + 7(10) = 100$
- B. $10n + 9n + 7n = 100$
- C. $4n + 10 = 100$
- D. $5n + 10 = 100$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

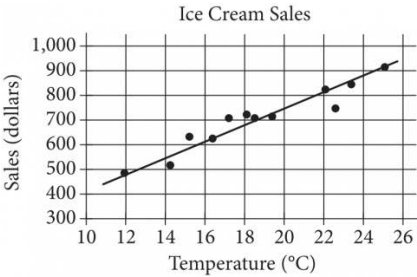
The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Hard

Question



The scatterplot above shows a company’s ice cream sales d , in dollars, and the high temperature t , in degrees Celsius ($^{\circ}\text{C}$), on 12 different days. A line of best fit for the data is also shown. Which of the following could be an equation of the line of best fit?

- Answer
- A. $d = 0.03t + 402$
 - B. $d = 10t + 402$
 - C. $d = 33t + 300$
 - D. $d = 33t + 84$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	p(x)
−2	5
−1	0
0	−3
1	−1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

Answer

- A. $(x - 3)$
- B. $(x + 3)$
- C. $(x - 1)(x + 2)$
- D. $(x + 1)(x - 2)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$x(r - 9) + 4 = 19x + 27$

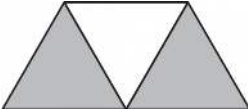
In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

Answer

- A. 4
- B. 9
- C. 23
- D. 28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

Answer

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

The density of a certain solid substance is **813** kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of **0.60** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

Answer

- A. 176
- B. 488
- C. 1,355
- D. 3,764

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
- II. $r(x) = a(4.1)^{x+b}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A train traveled **173** miles in the first **3** hours of a trip and traveled at an average speed of **53** miles per hour for the remainder of the trip. Which equation gives the total number of miles, ***y***, the train traveled if the trip was ***x*** hours long and ***x* > 3**?

Answer

- A. ***y* = 53*x* + 173**
- B. ***y* = 53*x* + 14**
- C. ***y* = 14*x* + 226**
- D. ***y* = 14*x* + 53**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is **972 cubic centimeters (cm^3)**, and the volume of prism Q is **36 cm^3** . One edge of prism P has a length of **12 cm**. What is the length, in **cm**, of the corresponding edge of prism Q?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

$11x + 14y \leq 115$

Anthony will spend at most \$115 to purchase x small cheese pizzas and y large cheese pizzas for a team dinner. The given inequality represents this situation. Which of the following is the best interpretation of $14y$ in this context?

Answer

- A. The amount, in dollars, Anthony will spend on each large cheese pizza
- B. The amount, in dollars, Anthony will spend on each small cheese pizza
- C. The total amount, in dollars, Anthony will spend on large cheese pizzas
- D. The total amount, in dollars, Anthony will spend on small cheese pizzas

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Module 14 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

21. _____

22. _____